

es, development centres and business process outsourcing (BPO) establishments in India. He says, "If you look at telecommunication as a segment, telecom network infrastructure roll-outs like 3G or 4G are the major driving factors for Telco." He adds, "With roll-outs, multinational companies come up with main switching centres, distribution centres or data centres that require power backup."

The broadening gap between demand and supply of power is another key factor for the growth of industrial inverters and UPSes. Fast-growing industrialisation, which has a positive aspect of the country's economic growth, comes with a difficult concern for the demand to keep pace with the growing infrastructure.

According to Malvika Sood, senior technical sales engineer, Texas Instruments India Pvt Ltd, the present inverter market in India produces about seven to eight million inverters per year and the growth is five to six per cent per year. She says, "Earlier the business was focused mainly on tier I and tier II cities. Now, with improved power situation in India, focus is also shifting to rural India." She adds, "As per McKinsey & Company's analysis report, total demand for power will rise from 120GW to 315GW by 2017. As the inverter industry thrives on shortage of power, this is the area where UPS markets have the greatest potential and see a speedy growth."

Today, the corporate sector's presence is not only felt in urban cities but is slowly trickling down into smaller towns as well, informs Sivarajan. He says, "The power scenario in developing cities and towns is not very promising." Das too feels that the increasing penetration of IT/ITes sector in tier II cities has boosted the UPS and inverter market.

Low-harmonics, high-efficiency and wide-power-range products in demand

With respect to the corporate and industrial sector, it is the high power range of UPS systems that primarily

experiences the maximum demand, informs Moumita Mukherjee, research analyst, Netscribes. There is also a high demand for systems where customers can assess the total cost of ownership, which helps them reduce additional expenses in the long run.

Rao says, "Low harmonics and high efficiency are some other sought-after specifications. UPSes and inverters that contribute to a clean environment are also in demand." He adds, "Gone are the days

Growth factors at a glance

- Power deficit and the consequent frequent power cuts
- Unstable power grid
- Rapid industrialisation
- Telecom network infrastructure roll-outs
- Penetration of IT/ITes sector in tier II cities

where customers looked for preventive maintenance. Today, customers expect vendors to give some element of predictability, with respect to harmonics, in UPSes and inverters."

The UPS market is slowly moving towards transformerless and modular UPS systems, notes Das. This is mainly due to space constraints and rise in Leadership in Energy & Environmental Design (LEED) certification in buildings. He says, "Initially, inverters were used to operate fans and lights but now these are designed to run desktops, air-conditioners and other household appliances."

The industry is also looking towards setting up offices and factories more efficiently. Sood says, "Products with high efficiency (about 80 to 90 per cent) are in demand, which is met by sine-wave inverters. Consumers are not only asking for higher-efficiency products but are looking to save power as well."

She adds, "There is also notable rise in demand for solar inverters, which not only cultivates natural resources during the day but also charges batteries to supply power at night. Lower-rating solar inverters, also called hybrid inverters (com-

bination of solar panel and inverter) of up to 500W or so, are available as catalogue products with tier I inverter manufactures." Higher-rating products are customised as per the requirement from the industry.

Multinationals dominate UPS segment, Indian companies have the edge over inverters

Analysts and industry experts feel, multinationals like Schneider Electric and others rule the UPS market in

India today and Indian players do not have a great market share. Indian companies are also trying to supply products with the same kind of reliability as required by corporates.

Inverter-operating conditions are slightly stringent, given the power condition in India. Products made locally are completely based on the experience that manufacturers have with local grids. "For instance, the grid in the USA or Europe is completely stable. Their inverter systems need not be as rugged as the ones made for Indian conditions, where the frequency could go below 45Hz or above 55Hz for a 50Hz grid," explains Sivarajan.

He adds, "Standard voltage of 230V in India can dip to as low as 110V or 120V. Therefore power backup systems for usage in India need to be designed keeping such conditions in mind. Indian companies like Su-Kam try to understand grid-voltage levels and other conditions in cities and villages, and design systems accordingly."

Market trends

In the recent past, there has been a rise in consumers' growing affinity towards high power, cost-effectiveness and energy-efficiency. Mukherjee says, "With a continual rise in awareness about eco-friendliness, Green UPS technology is gradually gaining momentum in India."

Customers are willing to experiment with new technologies. Rao informs, "For instance, in the IT